The Enviro Remote

IXXN341, Design For Health.

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Hello!

I'm Campbell Smith, in 2024 my dad was diagnosed with Motor Neuron Disease, he later passed away the 3rd of October 2025 due to the disease.

Motor Neuron Disease is a group of disorders that affect the bodies motor neurons, the nerves responsible for controlled muscle movement.

Quick facts about MND

Motor Neurone Disease NZ. (2026, February 12). Basic facts about MND. https://mnd.org.nz/about-mnd/what-is-mnd/basic-facts-about-mnd/

2-3 in 100,000 New Zealanders will be diagnosed with Motor Neuron Disease each year, higher than the global average.

Around 1 in 15,000 live with Motor Neuron Disease at any time.

Most live 20-48 months from first symptoms.

Motor Neuron disease has no cure.

About the Patient

Motor Neuron Disease Patients have little to no mobility, and become reliant on others and technology.

This can cause feelings around loss of independence and dignity for the patient.

It can put strain on close relationships, and makes caregiving a fulltime job often for partners and family.

Enviro Remote looks to take existing hardware and make a working remote app for Motor Neuron Disease patients to control their enviroment.

In New Zealand MND Patients are supplied with Tablets outfitted with eye gaze technology, and Augmented Communication software from Smartbox and Tobii Dynavox.



Smartbox Assistive Technology. (n.d.). Grid Pad 16 [Photograph]. Retrieved October 6, 2026, from https://smartboxaac.com/grid-pad/

The Current Grid software from Smartbox has options for patients to use IR sensors, programming and deploying them through hardware installed on newer devices, but is hidden in settings and doesn't show the patient the full capabilities of the technology

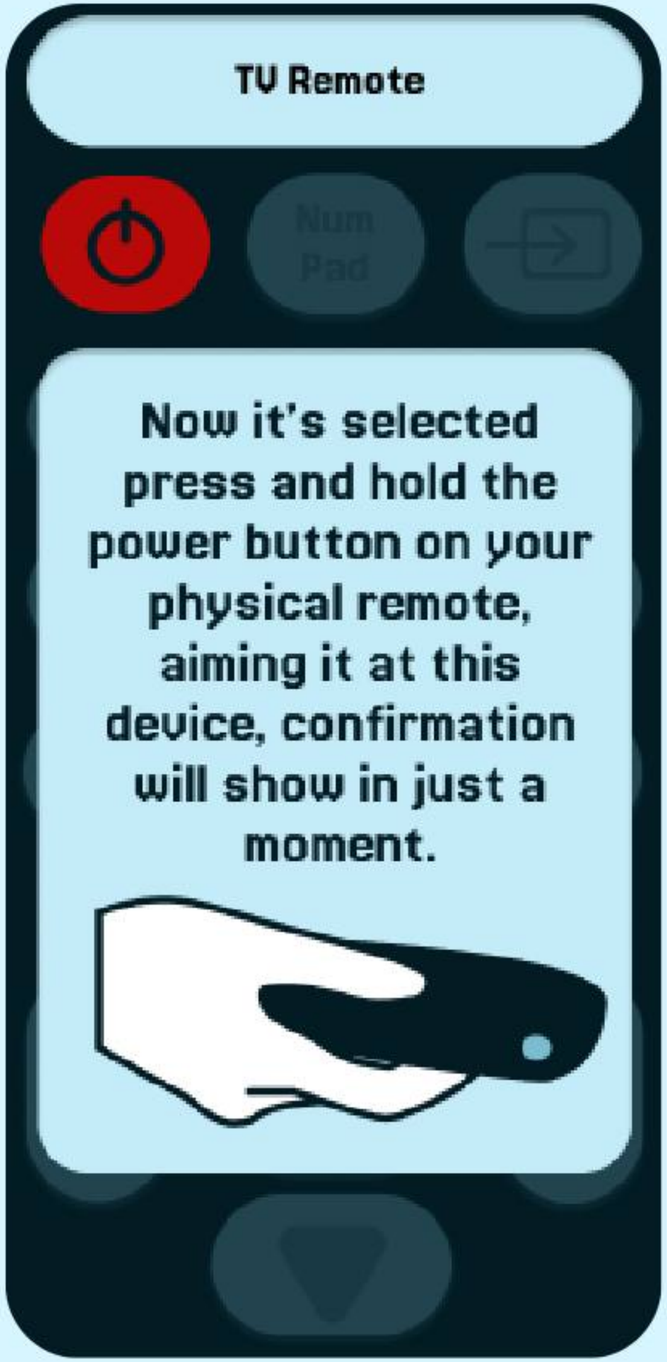


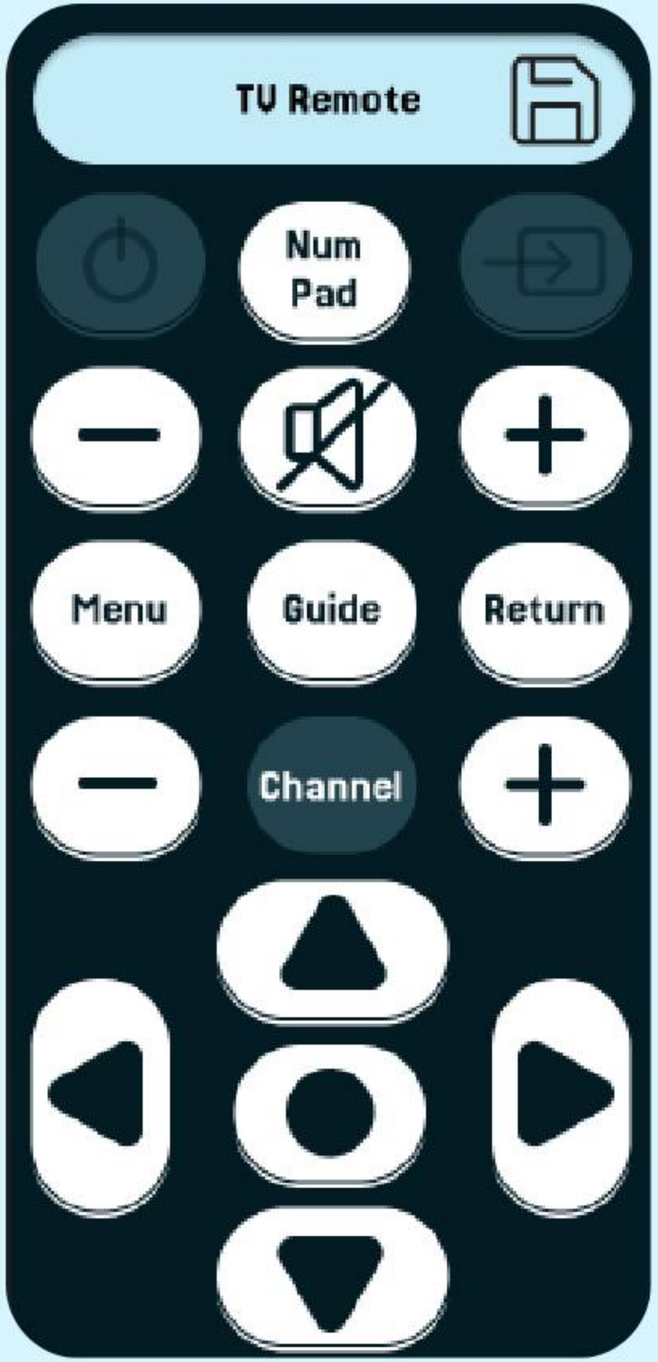
Smartbox Assistive Technology. (n.d.). Grid Pad 16 [Photograph]. Retrieved October 6, 2026, from https://smartboxaac.com/grid-pad/

This further limits the abilities of MND patients, putting a 'simple' task such as changing a TV channel behind a ICT barrier.



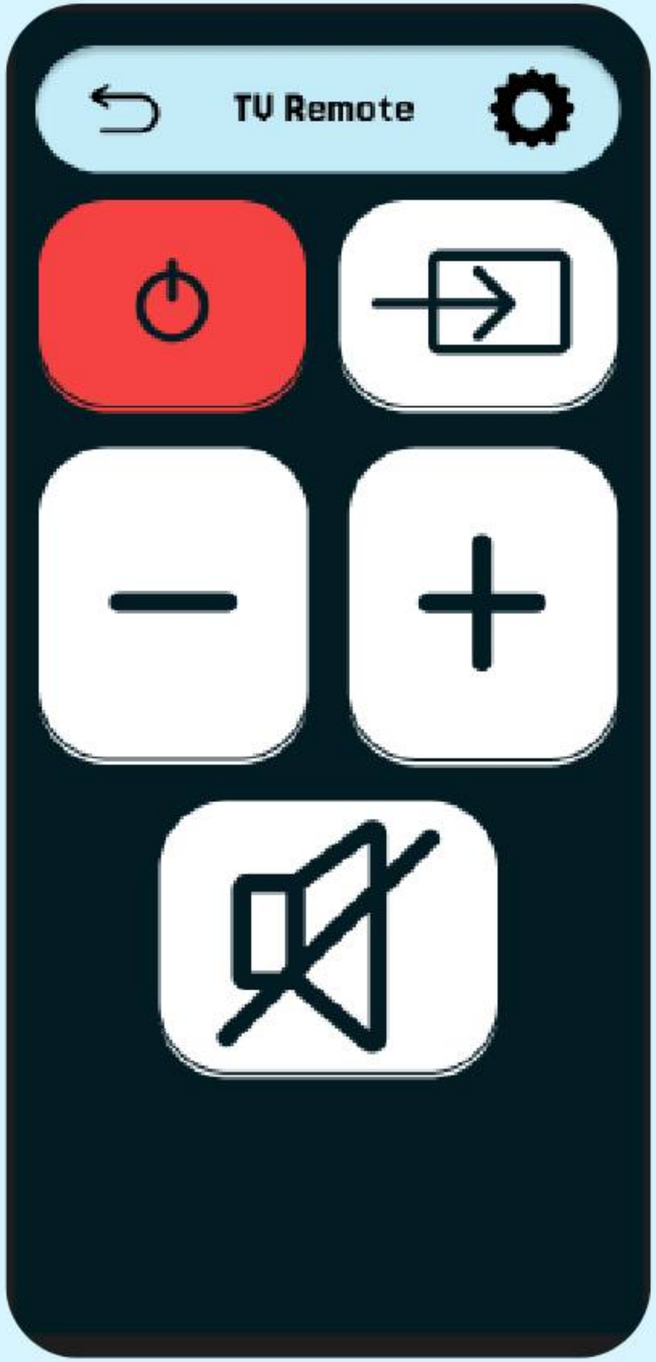
Tobii Dynavox. (n.d.). TD I-Series [Photograph]. Retrieved October 6, 2026, from https://www.tobiidynavox.com/pages/td-i-series

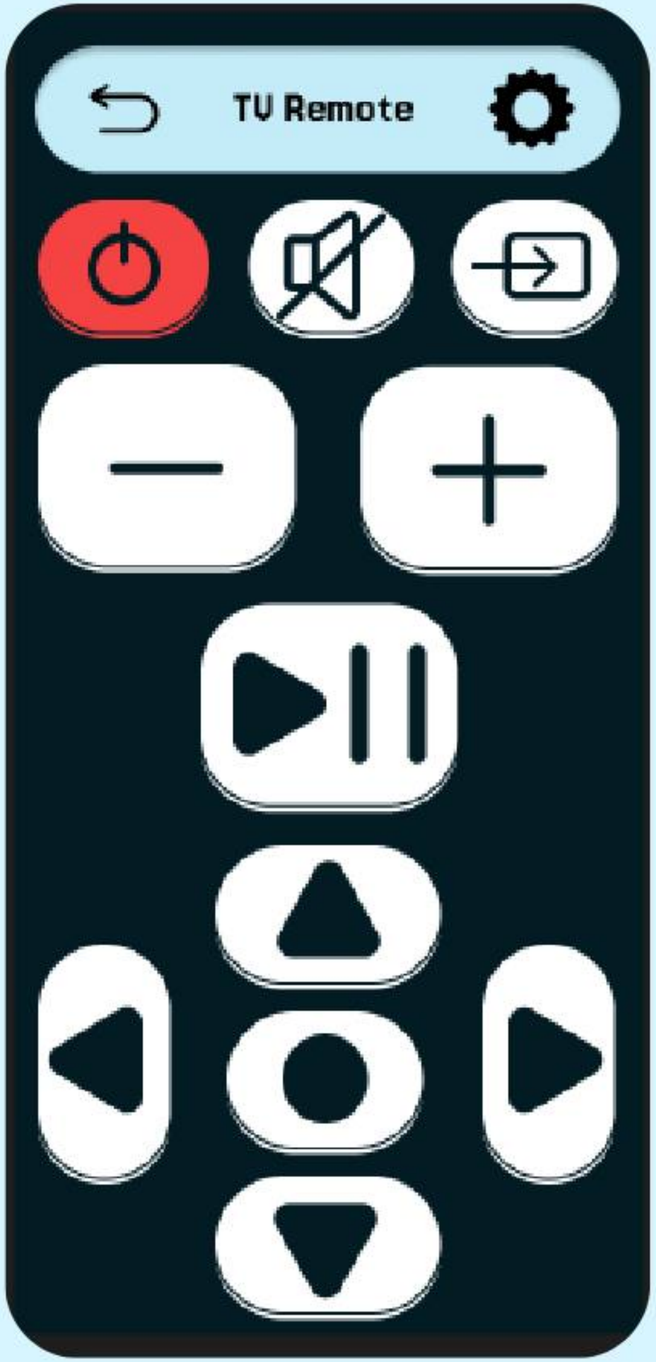




Enviro Remote looks to change this with its app.

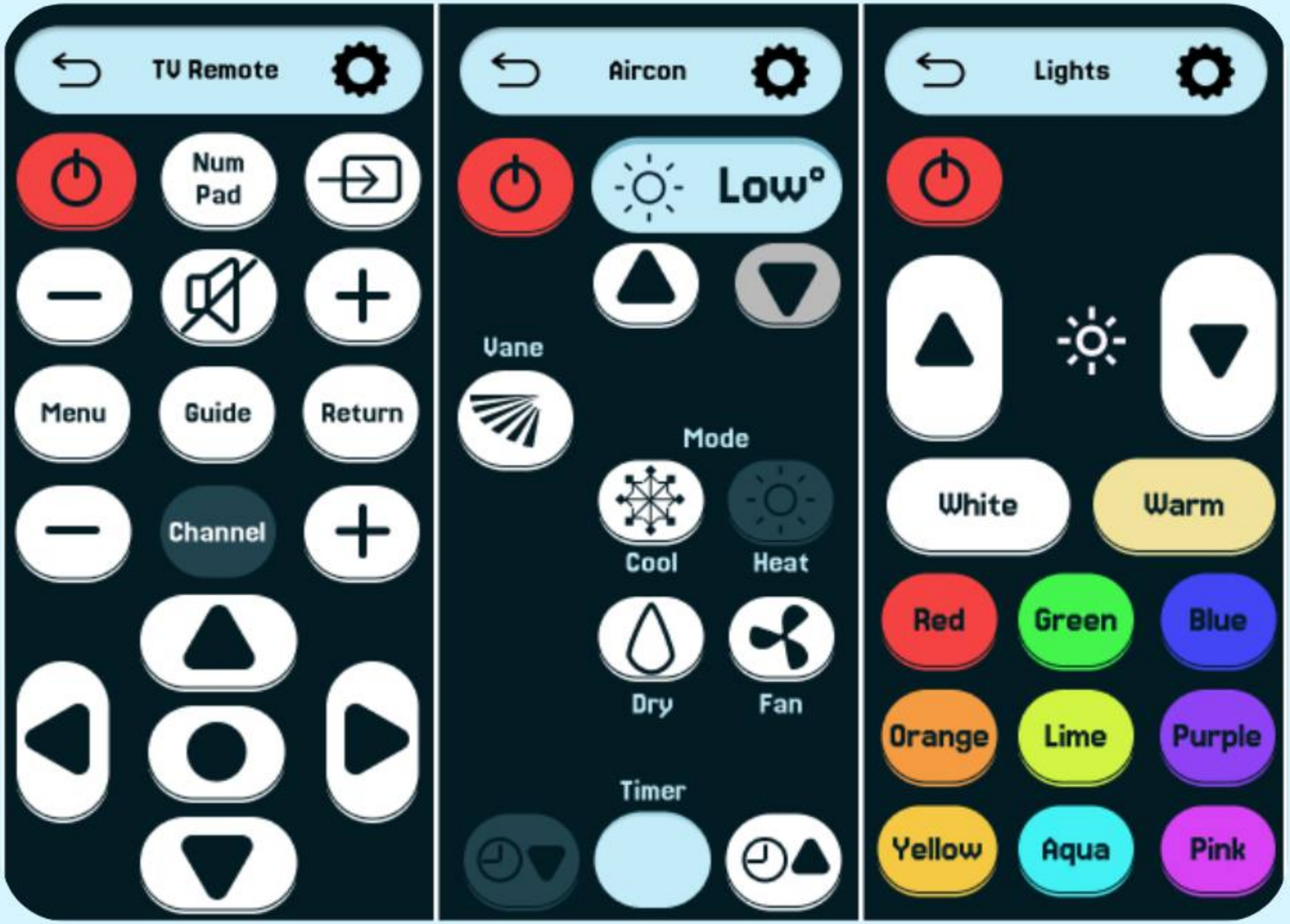
Putting a focus on onboarding for patients, and showing them what they really can do with their tech.





Enviro Remote offers a digital customisable remote, the user learns how to map and can then use to control their enviroment around them.

Anything and everything that uses a IR Remote can be mapped and used from the software, with presets for TV, Aircon, Lights and Call bells.



Enviro Remote faces the same challenges of the current smartbox solution, with IR mapping being a long system, but it takes the how to's and what can be mapped out of the patients worries, guiding them step by step through the process.

Risks

Complicated set up not suitable to non ICT Savy Users

Requires Physical Intervention for IR set-up

Time consuming, and involves mapping a lot of buttons

Mitigation

Revisitable onboarding to guide the users at any point

Onboarding is directed at users, and caregivers

Doesn't require all buttons to be mapped

Next Steps

Further premade layouts, and custom layout making, Remote layout making dashboard

User testing

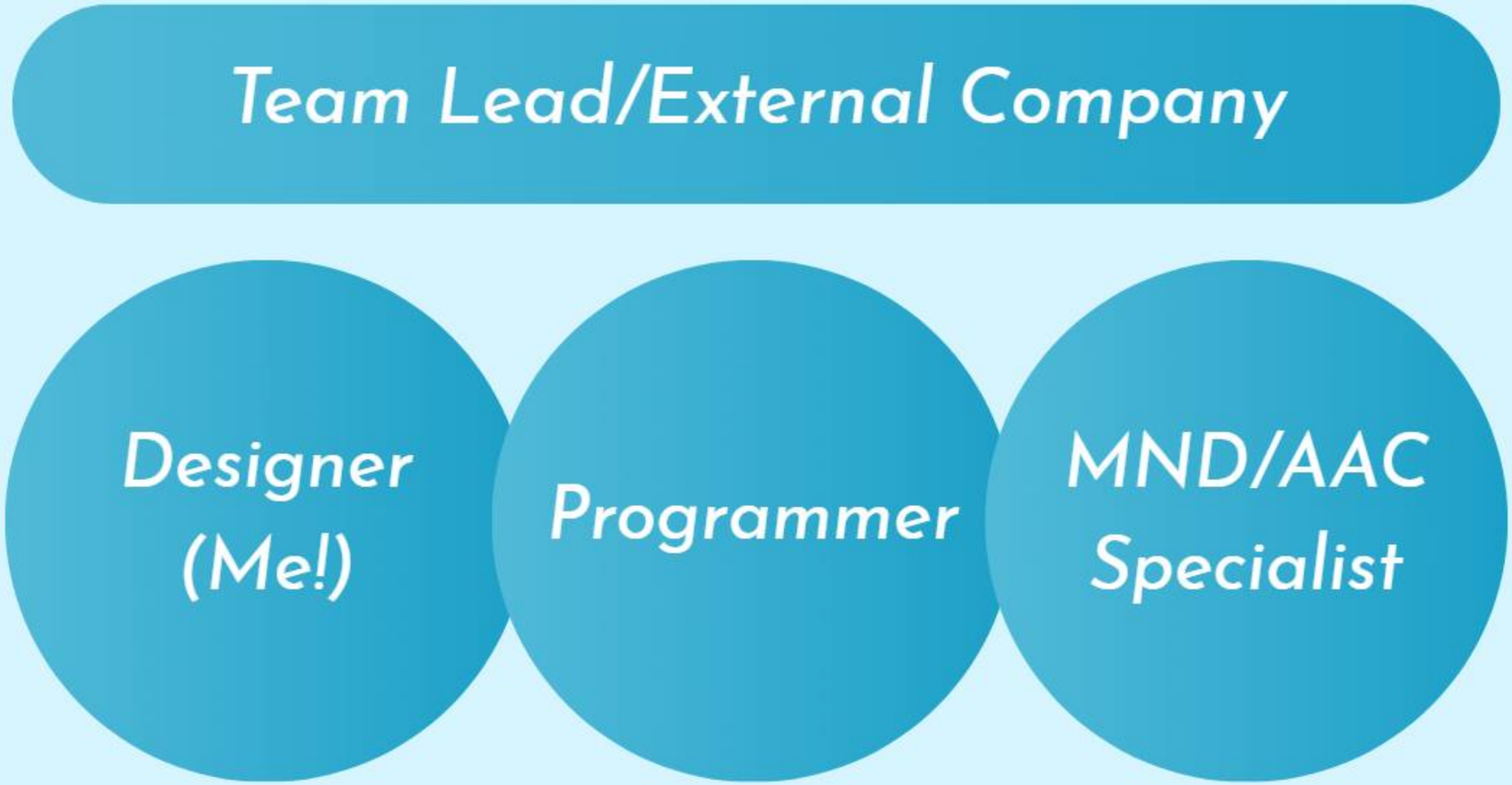
Common troubleshooting solutions for users to have a guide

Research into other IR controlled devices

Business Model

Enviro Remote Software

- Access Key Direct from Developer
  - *100% Of Profits for software to developers*
  - *Advertising, and marketing is self run and funded.*
  - *Lower cost for consumer through perchasing only their required software*
- Included with existing AAC software suites
  - *Percent from each software suite sale/subscription*
  - *Advertising, Marketing, and Medical relationships already built*


Team Lead/External Company
Designer (Me!)
Programmer
MND/AAC Specialist

Competition

SmartBox AAC Software

Risk of Competition for Enviro Remote is mitigated through pitching this software as an existing addition to existing AAC companies suite of softwares.

Tobii Dynavox

These companies have some level of existing solution, but lack insight and research in how to utilise the technology.

Summary

Enviro Remote is a software solution for MND Paitents to control their enviroment through IR sensors and recievers

MND Paitents have low to no physical mobility, relying on others to make changes to their enviroment, causing a lack of autonomy.

The technology already exists, but is under utilised by the companies in this space. Enviro Remote creates a user friendly solution that gives autonomy over paitents enviroments back.



Thank You

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